



Lora Based Coal Mine Safety Monitoring and Security System Using Rover

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ABSTRACT



Confined spaces often pose serious risks to workers due to the accumulation of toxic gases, insufficient oxygen levels, and extreme temperature or humidity conditions. To address these hazards, this project proposes a **Smart Safety System for Confined Spaces** that continuously monitors environmental parameters and transmits real-time data using LoRa communication technology. The system integrates a **gas sensor** for detecting harmful gases, a **DHT11 sensor** for temperature and humidity measurement, and an **oxygen sensor** for monitoring breathable air quality. An Arduino microcontroller collects sensor data and sends it wirelessly to a **Node MCU** module via LoRa for further processing and cloud integration. If any parameter exceeds safe limits, an audible **buzzer alert** is activated instantly to warn workers. The NodeMCU also uploads the data to an IoT dashboard for remote monitoring.

OBJECTIVE(S)



The primary objective of this project is to develop a smart safety system that ensures continuous monitoring of hazardous environmental conditions inside confined spaces. the goal is to create a portable, energy-efficient, and cost-effective solution that reduces the risk of accidents and ensures safe working conditions in environments such as manholes, tunnels, storage tanks, and industrial chambers.

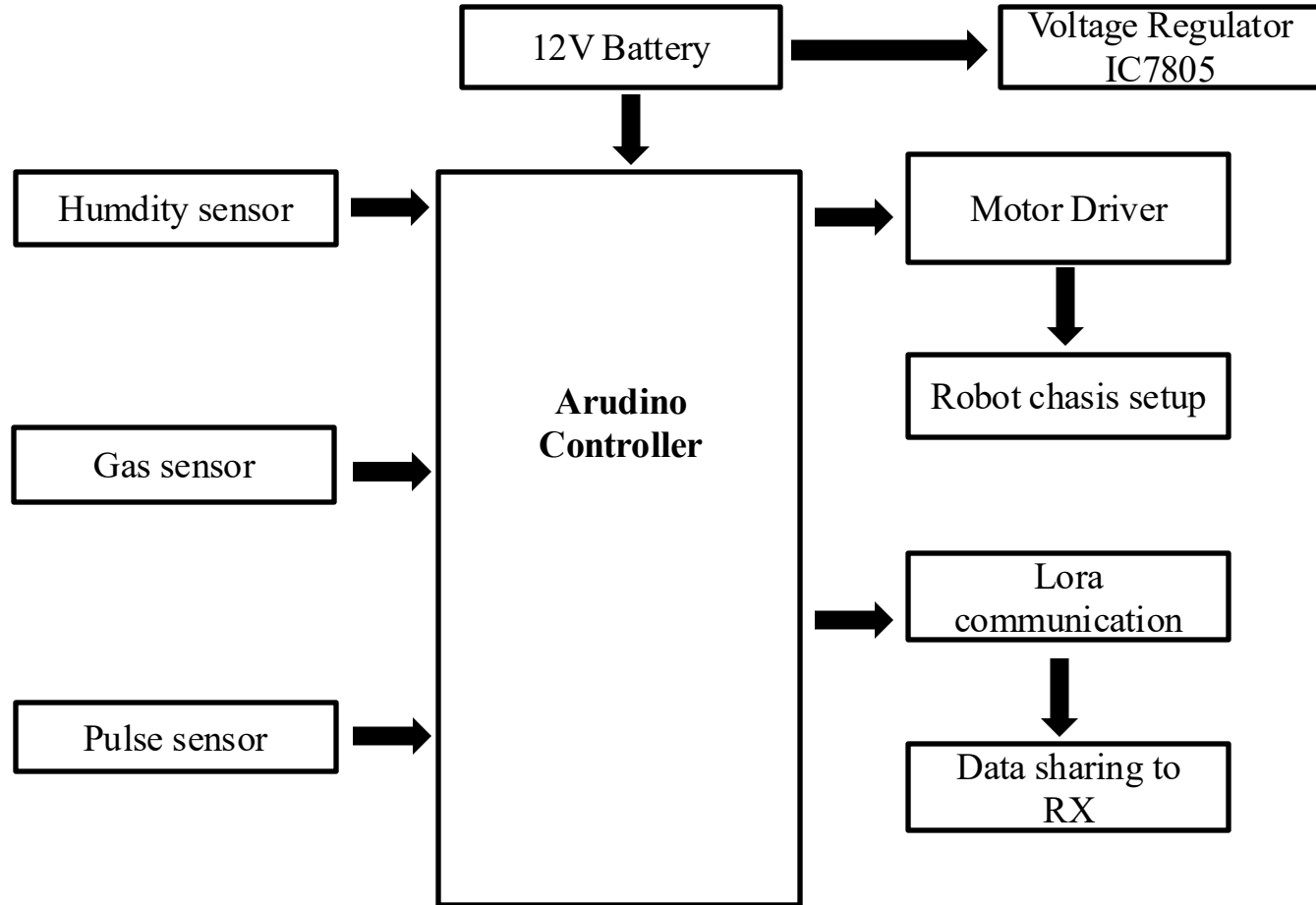
INTRODUCTION



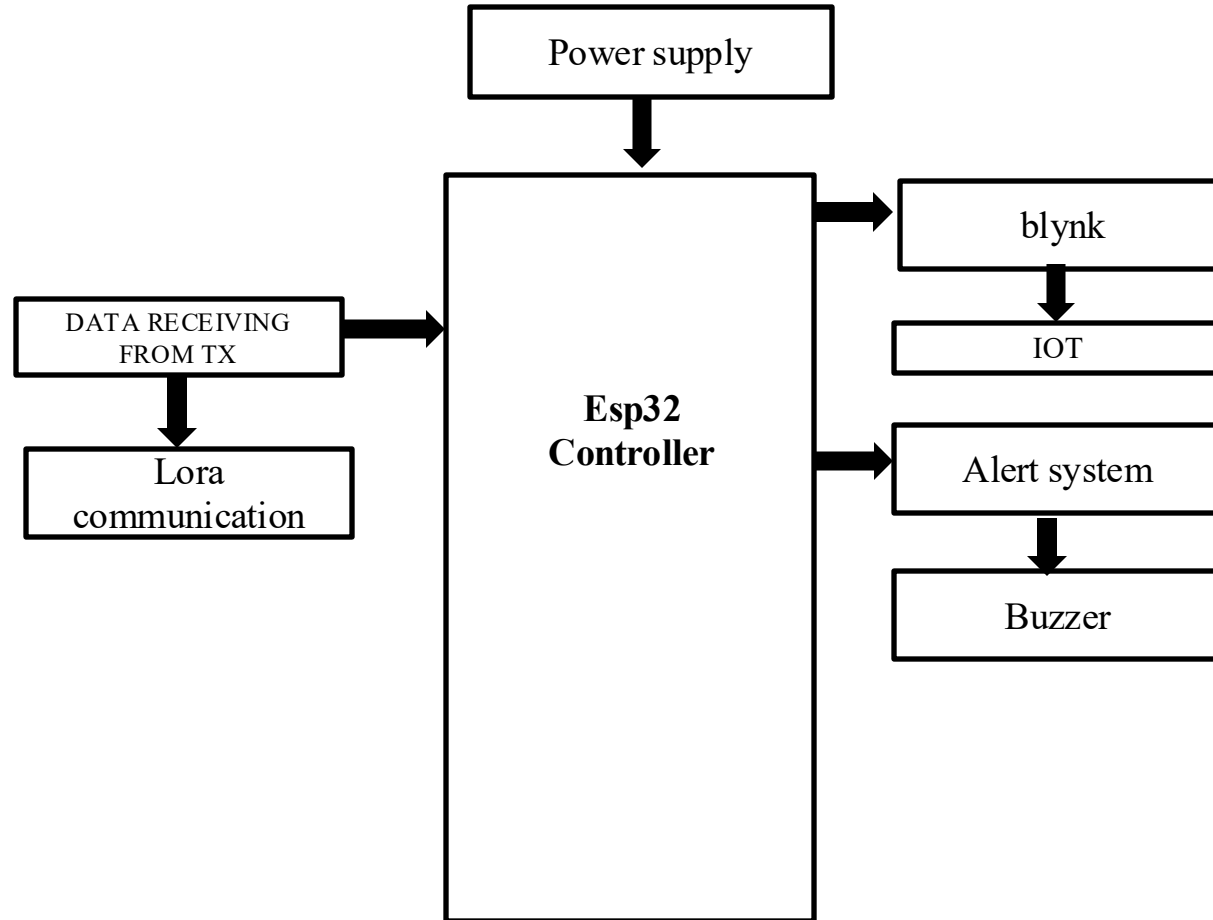
Coal mining is one of the most hazardous industries, where workers are exposed to dangerous conditions such as toxic gases, high temperatures, and the risk of explosions or collapses. Ensuring the safety and security of miners is therefore a critical challenge in modern mining operations. Traditional monitoring systems are often limited in range and real-time responsiveness, making it difficult to detect and react quickly to emergencies.

To address these issues, a **LoRa-based coal mine safety monitoring and security system using a rover** is proposed. This system utilizes long-range, low-power LoRa communication technology to continuously monitor environmental parameters such as gas levels, temperature, and humidity inside the mine. The rover acts as a mobile unit that can navigate through hazardous areas, collect data, and transmit it wirelessly to a remote monitoring station. This approach enhances real-time monitoring, improves worker safety, and enables quick emergency response in underground mining environments.

TX BLOCK DIAGRAM



RX BLOCK DIAGRAM



EXISTING SYSTEM



In the existing safety systems for confined spaces, most industries rely on handheld gas detectors and manual inspection methods to assess environmental conditions before allowing workers to enter. These traditional devices usually measure only a limited set of parameters such as toxic gases or oxygen levels and require workers to carry them at all times. Many systems lack continuous monitoring and do not provide real-time updates once the worker has entered the confined space. Additionally, alerts are often restricted to local indicators without remote communication, making it difficult for supervisors to monitor conditions from a safe distance. As a result, existing systems can be time-consuming, prone to human error, and inadequate for detecting sudden hazardous changes, which increases the risk of accidents in confined spaces.

PROPOSED SYSTEM



- The proposed system introduces an intelligent and automated safety solution designed to continuously monitor environmental conditions within confined spaces.
- It uses a combination of gas sensors, a DHT11 sensor, and an oxygen sensor to measure toxic gases, temperature, humidity, and oxygen levels in real time.
- An Arduino processes this data and transmits it wirelessly using LoRa technology to a NodeMCU module, enabling long-range and reliable communication.
- The NodeMCU uploads the data to an IoT dashboard, allowing supervisors to remotely track safety conditions. In case any parameter exceeds safe limits, the system immediately activates a buzzer to warn workers inside the confined space.
- This approach enhances safety by providing continuous monitoring, instant alerts, and remote visibility, making it a more advanced, efficient, and proactive alternative to traditional manual inspection methods.

MODULES



- **Sensor Module**

This module includes the gas sensor, DHT11 sensor, and oxygen sensor, which continuously measure toxic gases, temperature, humidity, and oxygen levels inside the confined space. The data is sent to the Arduino for processing.

- **Processing Module (Arduino)**

The Arduino receives the sensor data, analyzes the readings, compares them with predefined safety thresholds, and prepares the information for wireless transmission.

- **Communication Module (LoRa)**

The LoRa module enables long-range, low-power wireless communication. It sends the processed data from the Arduino to the NodeMCU module located outside the confined space.

- **Monitoring & IoT Module (NodeMCU)**

The NodeMCU receives the LoRa data and uploads it to an IoT dashboard or cloud server. Supervisors can remotely monitor real-time environmental conditions.

- **Alert Module (Buzzer)**

This module provides immediate on-site warnings. When the system detects unsafe gas levels, low oxygen, or abnormal temperature/humidity, the buzzer activates to alert workers instantly.

BATTERY

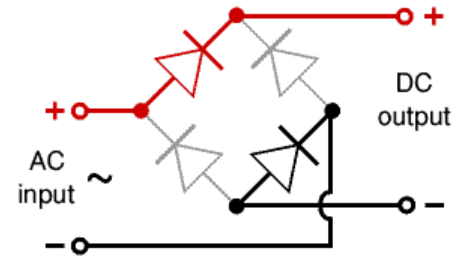
It provide the whole electricity to the project.

- If kept in a charged state when unused, the common lifespan of a 12-volt Gel or AGM battery.
- An average ambient temperature of 25 °C, the battery still retains 80 % of its original capacity.
- A battery uses one or more cells to allow a chemical reaction creating the flow of electrons in a circuit. Batteries do not create energy or power on their own.



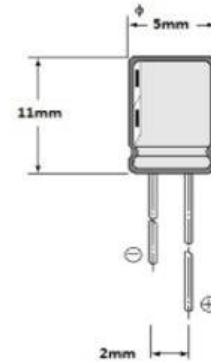
BRIDGE RECTIFIER

- O/p DC current is 1.5A
- The maximum peak reverse voltage is 800V
- Output Voltage: $(\sqrt{2} \times V_{RMS}) - 2 \text{ Volt}$
- The maximum input voltage is 560V
- Voltage drop for each bridge is 1V @ 1A
- The surge current is 50A



CAPACITOR

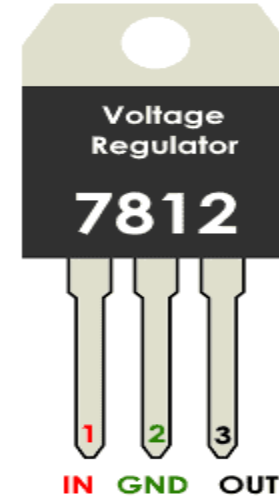
- **Capacitance:** 1000uF
- **Maximum Voltage:** 50 Volts
- **Tolerance:** $\pm 15\%$
- **Capacitor Type:** Radial Through Hole Electrolytic
- **Polarization:** Yes. Line with '-' denotes negative lead
- **Max Temperature:** $+85^{\circ}\text{C}$



VOLTAGE REGULATOR

LM7812

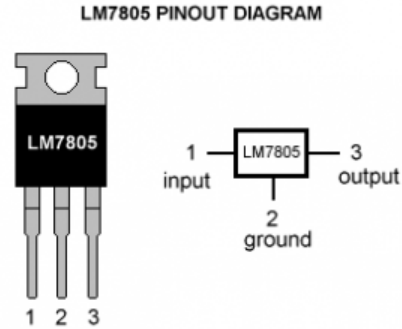
- Minimum Input voltage is 15V
- Maximum Input Voltage is 35V
- Current rating $I_c = 1A$
- Maximum Output Voltage $V_{Max}=12.5V$
- Minimum Output Voltage $V_{Min}=11.8V$



VOLTAGE REGULATOR

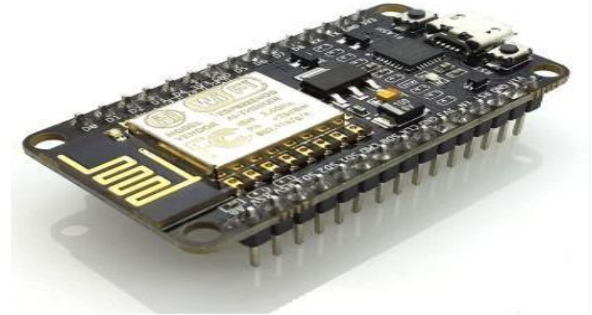
LM7805

- Minimum Input voltage is 7V
- Maximum Input Voltage is 35V
- Current rating $I_c = 1\text{A}$
- Maximum Output Voltage $V_{\text{Max}} = 5.2\text{V}$
- Minimum Output Voltage $V_{\text{Min}} = 4.8\text{V}$



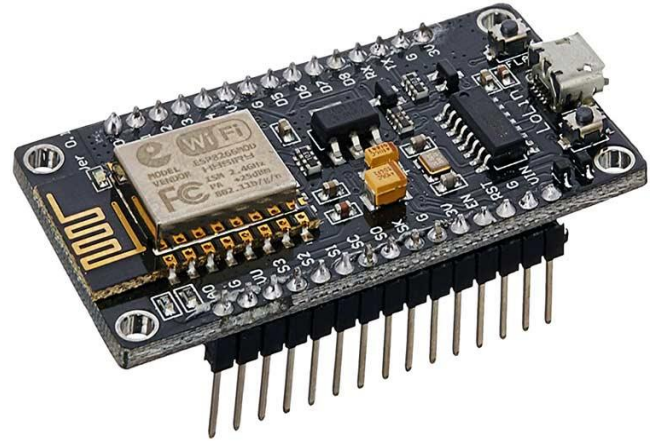
ESP32

- Microcontroller: ESP32 Development Board
- Operating Voltage: 3.3V DC
- Input Voltage (VIN): 5V to 12V DC
- Recommended Input Voltage: 7V to 12V DC
- Wi-Fi: 2.4 GHz Wi-Fi (802.11 b/g/n)
- Bluetooth: Bluetooth v4.2 (Classic + BLE)
- Digital I/O Pins: Up to 34 GPIO pins
- Analog Input Pins: Up to 18 ADC channels
- DAC Pins: 2
- PWM Channels: 16
- Flash Memory: 4 MB (typical)



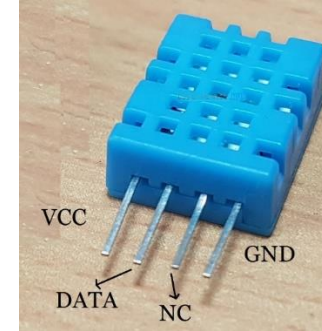
NODEMCU

- Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106
- Operating Voltage: 3.3V
- Input Voltage: 7-12V
- Digital I/O Pins (DIO): 16
- Analog Input Pins (ADC): 1
- UARTs: 1
- SPIs: 1
- I2Cs: 1
- Flash Memory: 4 MB
- SRAM: 64 KB
- Clock Speed: 80 MHz
- USB-TTL based on CP2102 is included onboard, Enabling Plug n Play
- PCB Antenna
- Small Sized module to fit smartly inside your IOT projects



DHT 11

- DHT11 Humidity Sensor consists of 4 pins: VCC, Data Out, Not Connected (NC) and GND.
- The range of voltage for VCC pin is 3.5V to 5.5V.
- A 5V supply would do fine.
- The data from the Data Out pin is a serial digital data.
- DHT11 Sensor can measure a humidity value in the range of 20 – 90% of Relative Humidity (RH) and a temperature in the range of 0 – 50°C.



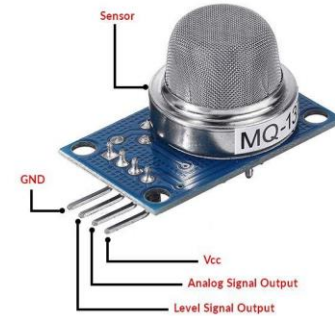
PULSE SENSOR

- This is a hear beat detecting and biometric pulse rate sensor
- Its diameter is 0.625
- Its thickness is 0.125
- The operating voltage is ranges +5V otherwise +3.3V
- This is a plug and play type sensor
- The current utilization is 4mA
- Includes the circuits like Amplification & Noise cancellation
- This pulse sensor is not approved by the FDA or medical. So it is used in student-level projects, not for the commercial purpose in health issues applications.



GAS SENSOR

- Operating Voltage is +5V
- Analog output voltage: 0V to 5V
- Digital Output Voltage: 0V or 5V (TTL Logic)
- Preheat duration 20 seconds
- Can be used as a Digital or analog sensor
- The Sensitivity of Digital pin can be varied using the potentiometer



BUZZER

- Rated Voltage: 6V DC
- Operating Voltage: 4-8V DC
- Rated current: <30mA
- Sound Type: Continuous Beep
- Resonant Frequency: ~2300 Hz
- Small and neat sealed package
- Breadboard and Perf board friendly



LORA MODULE

- The **433MHz SX1278 LoRa Module (Ra-02)** operates based on **LoRa (Long Range) spread spectrum modulation technology**, which is designed for long-range, low-power wireless communication.
- The module uses **LoRa (Long Range)** modulation, which is a type of **chirp spread spectrum (CSS)** modulation.
- LoRa allows signals to be spread over a wider bandwidth, increasing resistance to interference and allowing reception even below the noise floor.



ARDUINO IDE

- The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload programs and communicate with them.

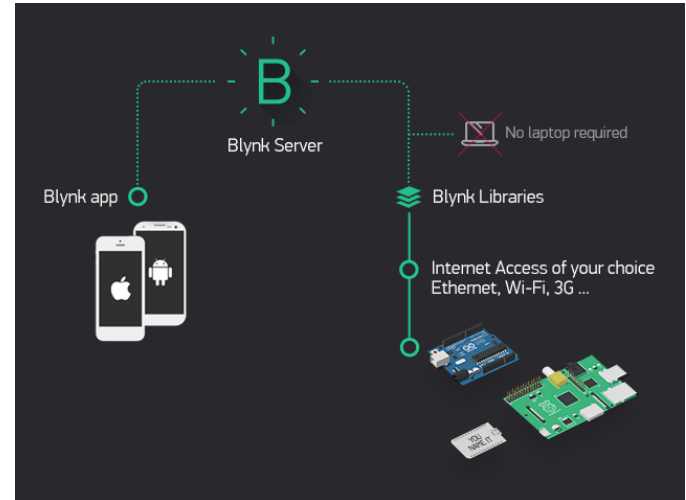


BLYNK APP

Blynk was designed for the Internet of Things. It can control hardware remotely, it can display sensor data, it can store data, visualize it and do many other cool things.

Features

- Similar API & UI for all supported hardware & devices
- Connection to the cloud using:
 - WiFi
 - Bluetooth and BLE
 - Ethernet
 - USB (Serial)
 - GSM
- Set of easy-to-use Widgets
- Direct pin manipulation with no code writing



ADVANTAGES



- **Continuous Monitoring:** The system checks the air conditions all the time, keeping workers safe.
- **Instant Alerts:** A buzzer warns immediately when the air becomes dangerous.
- **Long-Range Communication:** LoRa sends data far away, so supervisors can monitor from a distance.
- **Low Cost and Portable:** The device is affordable and easy to carry to any confined space.
- **Reduces Accidents:** Early warnings help prevent injuries and save lives.

APPLICATIONS



- **Manholes and Sewage Systems** – To monitor harmful gases and protect workers during maintenance.
- **Underground Tunnels** – For checking oxygen levels and temperature while workers operate below ground.
- **Storage Tanks** – Useful in chemical, oil, and water tanks where toxic gases may accumulate.
- **Industrial Chambers** – Ensures safety in boilers, reactors, and other enclosed industrial spaces.
- **Mining Areas** – Helps detect dangerous gas buildup and unsafe conditions underground.
- **Construction Sites** – For confined areas such as basements, pits, and enclosed structures.
- **Disaster Response** – Can be used by rescue teams to detect unsafe air conditions in collapsed buildings.

CONCLUSION

The Smart Safety System for Confined Spaces provides an efficient and reliable solution for protecting workers in hazardous environments. By integrating sensors with Arduino and LoRa communication, the system continuously monitors air quality and sends real-time data to the NodeMCU for remote supervision. Immediate alerts through a buzzer ensure quick action whenever unsafe conditions arise. This low-cost, portable, and easy-to-use system significantly reduces the risk of accidents by enabling early detection of dangerous gases and insufficient oxygen levels. Overall, the project enhances workplace safety and supports industries in maintaining safer operational practices within confined spaces.

FUTURE SCOPE

The Smart Safety System for Confined Spaces can be further improved by adding advanced features for better safety and reliability. In the future, the system can include GPS tracking to locate workers in emergencies, wearable devices for personal monitoring, and AI-based data analysis to predict dangerous conditions before they occur. The LoRa communication can be upgraded to LoRaWAN for wider network coverage and cloud integration. A mobile application can also be developed so supervisors can receive alerts and monitor data directly on their smartphones. Additionally, the system can be powered using rechargeable batteries or solar modules to make it more energy-efficient and suitable for long-term field use. These enhancements would make the system even more intelligent, user-friendly, and effective in ensuring worker safety in confined spaces.

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Thank
You